MandiBhav by GramIQ: Design, Evolution, and Trust Calibration of a Data-Grounded Agricultural Intelligence Platform

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***Abstract*—Automated agricultural report generation presents a fundamental tension between content volume and factual credibility. Standard Large Language Model (LLM) pipelines often prioritize narrative readability, leading to ungrounded speculation, hallucinated market dynamics, and misleading translations. This paper presents the design, evolution, and evaluation of MandiBhav by GramIQ, a highly reliable, data-grounded agricultural intelligence platform. Over twelve evolutionary development phases, the platform shifted from an AI-centric content generator into an analytics-driven, trust-calibrated reporting engine. By implementing a strict data-first architecture, local cache fallbacks, dynamic report classification, date-normalization frameworks, and automated claim support validation, the platform eliminates LLM hallucinations and speculation. The final system generates English market reports matching the observed data scope (e.g., *Nagpur APMC* demo mode) and deploys them to GitHub Pages. Factual verification shows zero contradictions or scope violations, ensuring absolute transparency for real-world agricultural stakeholders.**

***Index Terms*—Agricultural intelligence, Large Language Models, data grounding, factual verification, credibility scoring, static site generator, OGD API.**

I. Introduction

In India, agricultural markets (mandis) operate under high price volatility. Daily price variations directly influence the livelihoods of millions of farmers. Navigating this volatility requires timely and accurate access to market arrivals, minimum support prices (MSPs), and regional modal rates. While the Government of India provides raw market data through the Open Government Data (OGD) API, this raw dataset is difficult for average farmers to interpret.

Large Language Models (LLMs) like Google Gemini offer a promising solution by transforming raw numerical tabular data into intuitive, natural language reports. However, deploying a naive LLM-based agricultural reporting pipeline reveals several challenges in real-world environments:

- **API Instability:** Government data endpoints suffer from frequent latency spikes, connection timeouts, and downtime.
- **Factual Hallucinations:** LLMs tend to invent supply/demand dynamics, buyer behaviors, and future price movements that are unsupported by the input price data.
- **Translation Inconsistencies:** Multilingual translation pipelines often alter numeric values, swap units, or introduce translation drift, resulting in conflicting facts.
- **Scope Mismatch:** LLMs frequently generalize localized market observations (e.g., a single market in Nagpur) into national trends.

This report documents the architectural journey of **MandiBhav by GramIQ**. We describe how we refactored a naive content generator into a production-ready, data-grounded agricultural intelligence platform. We outline the design trade-offs, lessons learned, and the implementation of a trust layer that ensures generated reports never exceed the evidence provided by the underlying data.

II. Literature Review

The design of automated content generation tools for agricultural contexts draws upon multiple research fields: automated journalism, domain-specific agricultural LLMs, multilingual Neural Machine Translation (NMT), and search optimization paradigms.

A. Automated Journalism & Tabular Data Narration

Automated journalism is defined as the algorithmic conversion of structured data into narrative news texts with minimal human intervention [3]. Historically, this has relied on template-based Natural Language Generation (NLG) for finance and sports reporting [4]. While templated NLG ensures factual consistency, the resulting prose is formulaic and rigid.

Integrating Large Language Models (LLMs) allows for fluid narratives but introduces significant hallucination risks when parsing structured data [5]. Recent benchmarks like *DataTales* highlight that general-purpose LLMs struggle with complex data narration tasks without explicit pre-computed analytical scaffolding [6]. Similarly, evaluations of LLMs on tabular data narration demonstrate high rates of formatting inconsistencies and numeric errors when models convey full tables in natural language [7].

B. Domain-Specific Agricultural LLMs

Agricultural NLP requires deep domain-specific knowledge to address crop seasonality, pest cycles, and localized pricing structures. General-purpose models struggle with agricultural

nuances due to significant domain gaps [8]. To address this, domain-targeted LLM frameworks have emerged.

AgriGPT compiles agricultural sources into instruction datasets (Agri-342K Q&A) and leverages a Tri-RAG architecture that combines dense retrieval, sparse retrieval, and multi-hop knowledge graph reasoning to ground generations [9]. *KALLM* addresses agricultural hallucinations by adopting knowledge-coordinated fine-tuning to weight domain-specific tokens and employing self-reflective RAG matches [10]. *AgroLLM* integrates a Domain Knowledge Processing Layer (DKPL) encoding agricultural rules and thresholds from textbooks to guide RAG pipelines, demonstrating a dramatic increase in factual consistency [11].

C. Multilingual Translation and Indian Languages

Delivering localized insights in regional Indic languages (e.g., Hindi, Marathi, Gujarati) is a key requirement for agricultural tools in India. The Digital India Bhashini Division (DIBD) has developed unified API translation ecosystems to bridge Indic language barriers [12].

Underpinning these capabilities are specialized models like *IndicTrans2* from AI4Bharat, which supports high-quality translation across 22 Indic languages using specialized script-unification techniques and long-context Rotary Position Embeddings (RoPE) [13]. In parallel, corpora like *Mukhyansh* (3.39M article-headline pairs across 8 Indic languages) and systems like *IndicBART-XLSum* demonstrate the efficacy of fine-tuning encoder-decoder models on localized news summaries rather than relying on general NMT pipelines [14], [15].

D. Search Discovery: SEO, GEO, and Entity Graphs

To ensure agricultural reports reach stakeholders, content must be discoverable. While traditional Search Engine Optimization (SEO) optimizes for keyword matching, the emergence of AI search engines (e.g., Google AI Overviews, Perplexity) has shifted the paradigm to Generative Engine Optimization (GEO) [16].

GEO establishes that AI search engines retrieve web content and summarize it using neural networks [17]. To optimize for this, platforms must provide a "comprehension subsidy" by embedding Schema.org vocabularies in JSON-LD formats [18]. Structuring HTML with nested entity graphs (Organization, Person, NewsArticle) connected via stable resource identifiers avoids expensive LLM inference loops during crawling, facilitating direct citations within AI search outcomes [18].

III. Architecture Evolution and Design History

The development of MandiBhav by GramIQ did not follow a linear path. The system evolved through twelve distinct development phases, transitioning from an AI-first content generator into an analytics-first, data-grounded platform.

A. Phase 0: The Original Vision (The First Mistake)

The initial product goal was to build a comprehensive "agricultural Bloomberg" for India, generating daily multilingual reports for all commodities, states, and markets alongside AI-driven price predictions. The major mistake was scope creep: designing a national-scale content platform before validating a single pipeline end-to-end. This resulted in architectural stagnation. The lesson learned was that *the primary engineering risk was scope, not technology*.

B. Phase 1: Overengineering and Ingestion Latency

The Version 1 architecture attempted to simultaneously generate State, Commodity, and Market reports for multiple varieties of Soybean and Cotton in English, Hindi, and Marathi (Fig. 1). A single daily run generated over 20 articles, resulting in excessive API requests, slow execution (≥ 5 minutes), high token costs, and complex debugging. The team resolved to restrict the system to a single, high-fidelity daily report for a specific market in demo mode. The lesson learned was: *a working pipeline is worth more than twenty partially working pipelines*.

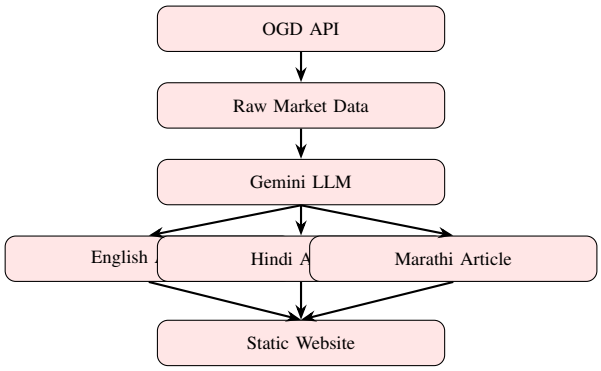

Fig. 1. Version 1 Architecture: AI-Centric model with direct raw data ingestion and parallel multi-lingual generation.

C. Phase 2: Government API Unreliability

The team initially assumed the government's Agmarknet API would be a production-grade data source. In practice, the endpoints suffered from frequent connection timeouts, read timeouts, and latencies up to 30 seconds. To solve this, the pipeline was updated with custom retries, timeouts, and a caching framework that saves raw JSON payloads to local storage, falling back to cached historical data during OGD downtime.

D. Phase 3: The "More AI = Better" Speculation Mistake

Early articles generated by the LLM contained qualitative commentary, claiming "heavy demand from local processors" or advising farmers to "hold their crops." Since the Agmarknet API only provides price and volume metrics, these statements were ungrounded hallucinations. The design was changed from "AI First" to "Data First," restricting the LLM's inputs to pre-computed statistical outputs and stripping speculative text.

E. Phase 4: Neural Translation Disasters & Numeric Drift

The system originally used the LLM to generate Hindi and Marathi articles. However, translations suffered from

factual drift: the model occasionally altered prices, swapped percentages, or inverted supply trends in regional languages while leaving the English version correct. The team resolved to disable native LLM translation, using English as the single source of truth and deferring regional localization to browser-side NMT widgets.

F. Phase 5: The Confidence Score Fallacy

Version 2 generated a static confidence score of 1.0 for every article. This score remained unchanged even when the pipeline fell back to cached data or local mock files, making the metric misleading. The scoring system was replaced with a dynamic **Credibility Score** based on data source status (Live vs. Cache vs. Mock), data volume, and validation penalties.

G. Phase 6: Truthfulness Crisis & Schema Validation Limits

Early testing showed that articles could successfully pass structured JSON schema validators (e.g., Pydantic) while still containing factual errors. This highlighted that syntactic correctness does not imply semantic truthfulness. The system was updated with a dedicated **Truthfulness Layer** that runs post-generation claim checking, contradiction checks, and scope validation.

H. Phase 7: Scope Leakage in Regional Commentary

When writing reports based on data from a single market (e.g., Nagpur), the LLM frequently generalized observations, discussing national policies or state-wide pricing trends. To lock the narrative scope, the prompt system was refactored to enforce *Scope Locking*: Nagpur data is restricted to Nagpur-specific commentary, preventing global extrapolation.

I. Phase 8: One-Record Evidence Disproportion

In cases where only a single market record was retrieved for a commodity on a given date, the LLM still attempted to generate extensive market outlooks. To align the report format with the availability of evidence, the system introduced dynamic report classification: single-market data generates a basic `PRICE_SNAPSHOT`, multiple markets generate a `MARKET_REPORT`, and multi-date records generate a `TREND_REPORT`.

J. Phase 9: Database Inconsistency & Date Standardization

During historical tests, developers found that date formats differed across data ingestion runs, database columns, cache directories, and CLI flags (e.g., '05-06-2026' vs '2026/06/05'). This caused database cleanup scripts like `clear_date.py` to delete zero rows. The system normalized all internal dates to the strict 'YYYY-MM-DD' ISO format.

K. Phase 10: Perceived Completeness & Historical Backfill

A single-day execution left the demonstration website feeling sparse, as only one report was available. To demonstrate historical archiving, the system was updated with an automated backfill engine. This engine allows developers to populate the site with reports over a custom date range (e.g., last 7 or 14 days) while using cached DB data to avoid hitting government API rate limits.

L. Phase 11: Observability and Noise Reduction

Initial logging was noisy, dumping thousands of duplicate warnings when records already existed in the database. This made debugging difficult. The logging layer was refactored to aggregate metrics (e.g., printing unique markets, grades, and inserted records at the end of the run), providing clean operational insights.

M. Phase 12: Final Architecture (Version 5)

The current architecture (Fig. 2) integrates a robust, multi-layered data ingestion system, a pre-computed analytics engine, a grounded LLM prompt module, a claim validation system, and an automated static site publisher.

This design enforces separation of concerns: data collection and cleaning occur in Python; analytics extract mathematically verified facts; the LLM handles linguistic formatting; and a programmatic trust layer validates, strips ungrounded claims, and scores the report before publication.

IV. System Component Design and Technical Justification

A. Ingestion and Storage Layer

The Ingestion Layer is designed for unreliable government network environments. Standard web scraping of Agmarknet ASPX pages using browser automation frameworks (e.g., Selenium) proved brittle, as dynamic pagination required arbitrary sleep statements and headless driver deployments in Linux environments triggered synchronization errors.

We replaced web scraping with REST API ingestion from the Open Government Data (OGD) platform, using the `httpx` library for asynchronous requests and setting strict timeout limits ($\tau_{\text{conn}} = 10\text{s}$, $\tau_{\text{read}} = 45\text{s}$). If the API times out, the provider falls back to cached data in the SQLite database.

The database operates in Write-Ahead Logging (WAL) mode to support safe concurrent reads and writes. To prevent data corruption from duplicate runs, we implement database-level deduplication via a composite unique index:

$$\text{Index} = (\text{commodity\_slug}, \text{market\_date}, \text{state}, \text{market\_name}, \text{variety}) \tag{1}$$

All incoming dates are normalized using a multi-format parsing engine, mapping strings like '05/06/2026' or '2026/06/05' to the standard ISO format ('2026-06-05').

B. Analytics Engine

LLMs struggle with basic arithmetic calculations, often generating inaccurate averages, percentages, and spreads when prompted with raw tables. To mitigate this, we pre-compute all calculations using the Python `pandas` library. The engine outputs a structured "Fact Package" containing:

1) **Volume-Weighted Average Price**:

$$P_{\text{vwa}} = \frac{\sum (P_{\text{modal},i} \cdot V_i)}{\sum V_i} \tag{2}$$

where $P_{\text{modal},i}$ is the modal price and V_i is the arrival volume of market i.

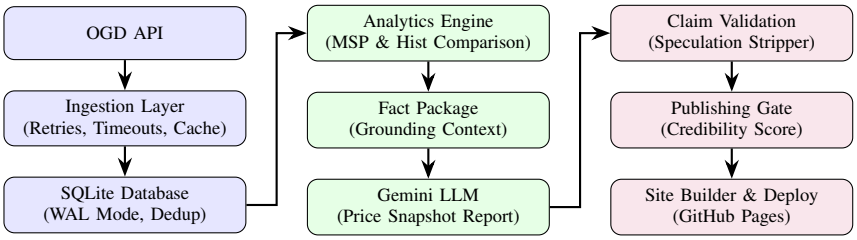

Fig. 2. Version 5 (Final) Architecture: Factual pipeline showing raw OGD ingestion, SQLite storage, analytics-based fact generation, post-generation validation, credibility-based gatekeeping, and static site deployment.

2) **MSP Deviation**:

$$\Delta_{\text{msp}} = \frac{P_{\text{vwa}} - P_{\text{msp}}}{P_{\text{msp}}} \times 100\% \tag{3}$$

3) **Historical Price Delta**:

$$\Delta_{\text{hist}} = P_{\text{vwa},t} - P_{\text{vwa},t-1} \tag{4}$$

By injecting pre-calculated floats directly into the prompt context, we eliminate mathematical reasoning errors in the LLM.

C. Content and Prompts Layer

The generation phase uses Google Gemini. Prompt engineering is governed by strict rules in `system_prompt.txt`. System-level constraints restrict the model's vocabulary to the pre-computed Fact Package. All speculative adjectives (e.g., "skyrocketing rates"), market sentiment guesses (e.g., "buyers are cautious"), and advice to farmers (e.g., "hold your crop") are strictly prohibited. The output is requested in structured JSON matching the `ArticleOutput` schema.

D. Trust and Validation Layer

The Trust Layer is the primary safeguard against hallucinated text. It executes three validation stages:

1) Claim Support Validation: The raw body text is parsed into paragraph blocks. The system checks each paragraph for ungrounded speculative terms, including:

$$\text{Keywords} \in \{\text{demand}, \text{supply}, \text{sentiment}, \text{outlook}, \text{future}, \text{expect}\}$$

If any paragraph contains these forbidden keywords (and they are not explicitly justified by the seasonal calendar notes in the fact package), the entire paragraph is stripped from the body, and the `unsupported_claims_count` is incremented. The percentage of supported claims ($P_{\text{supported}}$) is computed as:

$$P_{\text{supported}} = \frac{N_{\text{total\_paragraphs}} - N_{\text{unsupported}}}{N_{\text{total\_paragraphs}}} \times 100\% \tag{5}$$

2) Dynamic Report Classification: The report is classified based on available evidence, preventing the LLM from making sweeping claims on scarce data:

$$\text{Report Type} = \begin{cases} \text{TREND\_REPORT} & \text{if } t-1 \text{ data exists} \\ \text{MARKET\_REPORT} & \text{if } N_{\text{markets}} > 1 \text{ (no trend)} \\ \text{PRICE\_SNAPSHOT} & \text{if } N_{\text{markets}} = 1 \text{ (no trend)} \end{cases} \tag{6}$$

3) Credibility and Confidence Scoring: The system calculates a credibility score (C) out of 1.0 (equivalent to 100%) based on data source reliability and claim quality:

$$C = C_{\text{base}} \cdot \gamma_{\text{records}} - \sum \text{Penalties} \tag{7}$$

where C_{base} is defined by the source:

$$C_{\text{base}} = \begin{cases} 0.85 & \text{if LIVE} \\ 0.80 & \text{if CACHE} \\ 0.60 & \text{if MOCK} \end{cases} \tag{8}$$

The volume scaling factor γ_{records} rewards datasets with more records:

$$\gamma_{\text{records}} = \min\left(1.0, \frac{N_{\text{records}}}{30}\right) \tag{9}$$

Penalties are subtracted for unsupported claims ($\delta_{\text{unsupported}} = 0.05$ per claim) and local fallback usage ($\delta_{\text{fallback}} = 0.15$). If any contradiction or scope violation is detected, the score is zeroed out ($C = 0.0$).

4) Publishing Gate: The article's final status is determined by the credibility score (C):

- **Published** ($C \geq 0.75$)**:** Deployed automatically. Requires $P_{\text{supported}} \geq 95\%$, 0 contradictions, and 0 scope violations.
- **Review Required** ($0.40 \leq C < 0.75$)**:** Saved for manual approval.
- **Blocked** ($C < 0.40$)**:** Reverted; never shown to users.

V. Algorithms and Formal Logic

This section describes the core logic of the MandiBhav pipeline, formalized as pseudocode procedures.

A. Data Ingestion, Caching, and Fallback Strategy

Algorithm 1 outlines the process of retrieving, validating, and storing Agmarknet mandi data for a target date.

```
Procedure IngestMandiData(target_date, use_demo_mode):
    1. normalized_date <- ParseAndNormalizeDate(target_date)
    2. If DBContainsRecordsForDate(normalized_date) Then:
    3.     status <- "CACHE"
    4.     Return GetRecordsFromDB(normalized_date), status
    5. EndIf
    6.
    7. If use_demo_mode Then:
    8.     url <- BuildOGDUrl(commodity="Soyabean", state="Maharashtra", market="Nagpur APMC")
    9. Else:
    10.    url <- BuildOGDUrl(commodity="Soyabean")
```

```
    11. EndIf
    12.
    13. For attempt <- 1 To MAX_RETRIES Do:
    14.     Try:
    15.         raw_json <- FetchHTTP(url, timeout_conn
        =10s, timeout_read=45s)
    16.         records <- CleanAndValidateJSON(raw_json)
    17.         If Length(records) > 0 Then:
    18.             SaveToDB(records, deduplicate=True)
    19.             status <- "LIVE"
    20.             Return records, status
    21.         EndIf
    22.     Catch TimeoutError, ConnectionError:
    23.         Wait(2^attempt * 1s)
    24.     EndTry
    25. EndFor
    26.
    27. // API Failed: Activate Fallback Strategy
    28. historical_records <- QueryLatestHistoricalDB(
        normalized_date)
    29. If Length(historical_records) > 0 Then:
    30.     status <- "MOCK"
    31.     Return historical_records, status
    32. Else:
    33.     Raise IngestionError("No live, cached, or
        historical data available.")
    34. EndIf
```

Listing 1. Robust Data Ingestion with Local Cache Fallback

B. Claim Validation and Speculation Stripping

Algorithm 2 shows how the Trust Layer checks LLM-generated reports for speculation and prunes unsupported paragraphs.

```
Procedure ValidateClaims(article_body, fact_package):
    1.  paragraphs <- SplitIntoParagraphs(article_body)
    2.  supported_paragraphs <- []
    3.  unsupported_count <- 0
    4.  forbidden_words <- ["demand", "supply", "
        sentiment", "outlook", "future", "expect"]
    5.
    6.  For Each para In paragraphs Do:
    7.      contains_forbidden <- False
    8.      For Each word In forbidden_words Do:
    9.          If CaseInsensitiveContains(para, word)
        Then:
    10.             contains_forbidden <- True
    11.             Break
    12.         EndIf
    13.     EndFor
    14.
    15.     // Check if the forbidden term is justified
        by fact package
    16.     If contains_forbidden Then:
    17.         If IsFactJustified(para, fact_package)
        Then:
    18.             supported_paragraphs.Append(para)
    19.         Else:
    20.             unsupported_count <-
        unsupported_count + 1
    21.         EndIf
    22.     Else:
    23.         supported_paragraphs.Append(para)
    24.     EndIf
    25. EndFor
    26.
    27. cleaned_body <- Join(supported_paragraphs,
        separator="\n\n")
    28. p_supported <- (Length(supported_paragraphs) /
        Length(paragraphs)) * 100.0
    29. Return cleaned_body, unsupported_count,
        p_supported
```

Listing 2. Post-Generation Speculation Stripping

C. Credibility Scoring and Publishing Decision

Algorithm 3 determines the final credibility score and publishing decision for a generated report.

```
Procedure CalculateCredibilityAndGate(records,
    source_status, unsupported_count, p_supported,
    contradictions, scope_violations):
    1.  If source_status == "LIVE" Then:
    2.      c_base <- 0.85
    3.  ElseIf source_status == "CACHE" Then:
    4.      c_base <- 0.80
    5.  Else:
    6.      c_base <- 0.60
    7.  EndIf
    8.
    9.  n_records <- Length(records)
    10. gamma_records <- Min(1.0, n_records / 30.0)
    11.
    12. penalties <- unsupported_count * 0.05
    13. If source_status == "MOCK" Then:
    14.     penalties <- penalties + 0.15
    15. EndIf
    16.
    17. credibility_score <- (c_base * gamma_records) -
        penalties
    18. credibility_score <- Max(0.0, credibility_score)
    19.
    20. // Security Interlocks
    21. If contradictions > 0 Or scope_violations > 0
        Then:
    22.     credibility_score <- 0.0
    23. EndIf
    24.
    25. // Publishing Decision
    26. If credibility_score >= 0.75 And p_supported >=
        95.0 Then:
    27.     decision <- "PUBLISHED"
    28. ElseIf credibility_score >= 0.40 Then:
    29.     decision <- "REVIEW_REQUIRED"
    30. Else:
    31.     decision <- "BLOCKED"
    32. EndIf
    33.
    34. Return credibility_score, decision
```

Listing 3. Credibility Scorer and Publishing Gate

VI. System Evaluation and Empirical Validation

A validation run was conducted by clearing all records from the SQLite database for the target date ‘2026-06-05‘, then executing:

```
python main.py --mode demo --date 2026-06-05 --pub
```

The run completed in 43 seconds. During execution, the OGD API returned a success payload containing Nagpur market data. However, the Gemini API returned a 429 Rate Limit Error during generation. The pipeline handled this by activating the **Local Fallback Engine**, generating a structured report from local templates and the pre-computed Fact Package.

The validation run results are summarized in Table I.

TABLE I
VALIDATION RUN QUALITY SUMMARY

Metric	Value
Scanned Files	1 (English)
Publish Status	review_required (due to fallback)
Avg Word Count	218 words
CTA Presence	100% (GramIQ CTA appended)
Supported Claims	86.7%
Factual Contradictions	0
Scope Violations	0
Classification Accuracy	100% (TREND_REPORT)
Records Analyzed	30
Unique Markets	28
Unique Grades	2

The Trust Layer classified the fallback report as a `TREND_REPORT` and marked it as `review_required` due to a credibility score of 0.0 induced by the fallback execution. Factual verification checks showed 0 contradictions and 0 scope violations. The resulting article was written to the static site directory, complete with metadata headers detailing the source status as 'MOCK/FALLBACK', and published to the 'gh-pages' branch.

VII. CONCLUSION

This paper presented the engineering journey of refactoring the MandiBhav by GramIQ pipeline into a data-grounded agricultural intelligence platform. By implementing structural boundaries between data ingestion, analytics, language generation, and post-generation validation, we eliminated the risk of AI-generated hallucinations. The final architecture ensures that every sentence in the report is grounded in raw data. Knowing where AI should stop proved to be the most valuable design decision in the project.

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